

SAGARIKA RAMESH

(401) 499-9856 | sagarikaramesh251@gmail.com

<https://sagarikaramesh.com/> | <https://www.linkedin.com/in/sagarika-ramesh> | <https://github.com/Sagarika-Ramesh>

TECHNICAL SKILLS

ML & AI: Supervised Learning, Time Series Forecasting, Deep Learning, NLP, Anomaly Detection, Ensemble Methods, RAG, LLM Fine-tuning, Prompt Engineering, Gen AI, Agentic Workflows, SHAP Explainability

Languages & Tools: Python, SQL, R, JavaScript, MATLAB, PostgreSQL, Neo4j, pgvector, Cognite, Tesseract OCR, PySpark, PyTorch, TensorFlow, Pandas, NumPy

Cloud & Analytics: AWS Bedrock, Azure, GCP BigQuery, Tableau, Google Analytics, Looker Studio

PROFESSIONAL EXPERIENCE

IQumulus (Data Science Consulting firm) | Senior Data Scientist

2024 - Present

Transocean - Treasury Quant Forecasting

- Built a 13-week AP and AR cash flow forecast model using predictive modeling techniques, reducing quarterly AP forecast error by \$27M and AR forecast error by \$121M against baseline, a 31% improvement in collections forecast accuracy.
- Delivered AP and AR model evaluation summaries translating complex outputs into actionable Excel based SME deliverables for treasury stakeholders.

Transocean - Executive Dashboard

- Sole data scientist responsible for building an executive KPI dashboard used daily in operational standups, consolidating lagging and leading Financial, Safety, and Operational performance metrics across the global drilling fleet using advanced analytics and agile delivery.

IQumulus R&D - Subsea Intelligence Platform

- Designed a multi-agent Gen AI platform for subsea oil & gas using Claude API, LangChain, and MCP, combining RAG, NL-to-SQL, and a P&ID Intelligence Pipeline to extract and query structured and unstructured engineering data from complex diagrams.
- Enriched domain knowledge base with subsea engineering literature and P&ID diagrams using Voyage AI embeddings, enabling natural language access to specialized operational knowledge.

PC Classic Cars - Intelligent CRM & Inventory Tracking

- Built a LLM powered agent using Claude and PostgreSQL that parsed customer emails and phone transcripts, extracted preferences and intent, and autonomously classified inquiries as qualified leads, routing confirmed leads into a structured CRM database and triggering the sales pipeline.
- Built inventory and sales dashboards tracking order trajectory, inventory flow across 60+ units averaging 34 days, and flagging aged or low profit vehicles; integrated Google Analytics and Looker Studio for marketing tracking.

HESS - Executive Dashboard

- Built a custom executive dashboard tracking KPIs across Safety, Reliability, Production, Cost, and People, pulling live data via ETL pipelines from Cognite and Synergi across 3 Gulf of Mexico offshore assets.
- Consolidated real-time production, safety incidents, equipment vulnerability, and flow assurance sensor data into a single executive view, saving leadership 20+ hours of manual reporting per month.

EWS - Renewable Energy Forecasting

- Forecasted solar power output for 5 ERCOT solar plants using statistical analysis, comprehensive atmospheric and radiation features with cyclic feature engineering, achieving a prediction error of just 4.5 MW. Selected XGBoost over LSTM for robustness to 60 day data lag and irregular gaps.
- Validated model using SHAP analysis confirming it learned true physical drivers of solar generation, making results explainable to operations stakeholders.

Extreme Networks | ML Engineer

Jun 2019 - Jun 2021

- Built an end-to-end Widget Detection Model using Haar Cascade Classifier to detect widgets representing device health in Extreme CloudIQ, achieving 95% accuracy.
- Designed and developed 50+ automated test cases using Python and Selenium for Extreme CloudIQ as part of a CI/CD Automation Framework Development initiative.

EDUCATION

Brown University

M.S. Data Science | GPA: 3.93 / 4.0

Providence, RI

2021 - 2023

Sri Jayachamarajendra College of Engineering

B.E. Computer Science & Engineering | GPA: 3.56 / 4.0

Mysore, India

2015 - 2019